

MBAS Installation Requirements & Setup Guide

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Target Audience: IT Administrators, System Installers, Technical Support

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System Requirements

Minimum Requirements

Component	Specification
Operating System	Windows 10 (64-bit) or later
Processor	Intel Core i3 or AMD Ryzen 3 (2.0 GHz+)
RAM	4 GB
Storage	2 GB available disk space
Display	1366 x 768 resolution
Network	Not required (offline-first system)

Recommended Requirements

Component	Specification
Operating System	Windows 11 (64-bit)
Processor	Intel Core i5 or AMD Ryzen 5 (2.5 GHz+)
RAM	8 GB or more

Component	Specification
Storage	5 GB SSD storage
Display	1920 x 1080 resolution or higher
Network	Internet connection for initial setup only

Database Capacity

- **SQLite Database:** Supports 100,000+ records
- **Concurrent Users:** Up to 5 simultaneous connections
- **Transaction Volume:** 500+ invoices per day

Pre-Installation Checklist

Before installing MBAS, verify the following:

Software Prerequisites

- ☐ **Python 3.11 or 3.12** installed (see [Required Software](#))
- ☐ **SQLite3** included with Python (verify with `sqlite3 --version`)
- ☐ **Microsoft Visual C++ Redistributable** (2015-2022) installed
- ☐ **Windows PowerShell 5.1** or later (included in Windows 10/11)
- ☐ **Modern Web Browser** (Chrome, Edge, Firefox - latest version)

System Configuration

- ☐ **Administrator privileges** available for installation
- ☐ **User Account Control (UAC)** enabled
- ☐ **Windows Defender** or antivirus exceptions configured (if needed)
- ☐ **Firewall** configured to allow localhost connections
- ☐ **Disk space** verified (minimum 2 GB free)

Network Requirements

- ☐ **Internet connection** for initial dependency installation (one-time)
- ☐ **Localhost access** not blocked by security software
- ☐ **Ports 8000 and 5173** available (or custom ports configured)

Business Readiness

- ☐ **Business information** ready (name, address, tax ID, currency)
- ☐ **Logo image** prepared (PNG/JPG, recommended 200x200px)
- ☐ **User accounts** planned (Admin, Manager, Sales roles)

☐ **Product list** prepared for import (optional)

Required Software

1. Python Installation

Version Required: Python 3.11.x or 3.12.x

Download Python

1. Visit: <https://www.python.org/downloads/>
2. Download **Python 3.12.x** (latest stable release)
3. Choose **Windows installer (64-bit)**

Installation Steps

1. Run the Python installer
2.  **IMPORTANT:** Check "Add Python to PATH"
3.  Check "Install pip"
4. Click "Install Now"
5. Wait for installation to complete
6. Click "Close"

Verify Installation

Open Command Prompt and run:

```
python --version
```

Expected output:

```
Python 3.12.x
```

Check pip installation:

```
pip --version
```

Expected output:

```
pip 24.x.x from ... (python 3.12)
```

2. SQLite3 (Included with Python)

SQLite3 is automatically included with Python. Verify installation:

```
python -c "import sqlite3; print(sqlite3.sqlite_version)"
```

Expected output:

```
3.45.x
```

3. Microsoft Visual C++ Redistributable

Required for: Python cryptography packages

Download & Install

1. Visit: <https://learn.microsoft.com/en-us/cpp/windows/latest-supported-vc-redist>
2. Download: **vc_redist.x64.exe** (64-bit version)
3. Run installer
4. Accept license and click "Install"
5. Restart computer if prompted

4. Web Browser

Supported Browsers:

- Google Chrome (recommended)
- Microsoft Edge
- Mozilla Firefox

Minimum Version: Any version from 2024 or later



Installation Steps

Method 1: Production Installation (Recommended)

This method uses the pre-packaged deployment ZIP file.

Step 1: Extract Package

1. Download `MBAS_v1.0.7_Basic_20260427_DevOps.zip`
2. Right-click → "Extract All..."
3. Choose installation directory (e.g., `C:\MBAS` or `D:\MBAS`)
4. Click "Extract"

Recommended Paths:

- `C:\MBAS` (system-wide installation)

- `D:\MBAS` (separate drive installation)
- `F:\Applications\MBAS` (network drive - ensure fast access)

✗ Avoid Paths With:

- Spaces (e.g., "C:\Program Files\MBAS")
- Special characters (e.g., "C:\Apps&Tools\MBAS")
- Non-English characters
- Very long paths (> 100 characters)

Step 2: Run Installer

1. Navigate to extracted folder
2. Double-click `INSTALL.bat`
3. Wait for installation (2-5 minutes)

What Happens During Installation:

- Creates Python virtual environment (`venv/`)
- Downloads and installs dependencies
- Initializes SQLite database
- Creates desktop shortcut "MBAS"
- Configures system tray launcher

Step 3: Wait for Completion

Installation progress:

```
[1/5] Creating virtual environment...
[2/5] Installing backend dependencies...
[3/5] Installing system tray components...
[4/5] Initializing database...
[5/5] Creating desktop shortcut...
```

Installation complete! 

Total Time: 2-5 minutes (depends on internet speed)

Method 2: Development Installation

For developers or custom installations.

Step 1: Clone or Extract Source Code

```
cd D:\
git clone https://github.com/your-repo/mbas.git
cd mbas
```

Or extract source ZIP to desired location.

Step 2: Create Virtual Environment

```
python -m venv venv
venv\Scripts\activate
```

Step 3: Install Backend Dependencies

```
cd backend
pip install -r requirements-lock.txt
```

Step 4: Install Frontend Dependencies

```
cd ../frontend
npm install
```

Step 5: Initialize Database

```
cd ../backend
python src/scripts/init_db.py
```

Step 6: Create Admin User

```
python -c "from backend.src.scripts.init_db import create_default_admin;
create_default_admin()"
```



First-Time Setup

Launch MBAS

Method 1: Desktop Shortcut (Recommended)

1. Double-click "**MBAS**" icon on desktop
2. Wait 3-5 seconds (first-time dependency setup)
3. System tray icon appears (green checkmark)
4. Browser opens automatically to <http://localhost:8000>

First Launch Behavior:

- First click: 3-5 second delay (one-time system tray setup)
- Subsequent launches: Instant (< 1 second)

Method 2: Manual Launcher

1. Navigate to MBAS installation folder
2. Double-click `START_MBAS_TRAY.bat`
3. Same behavior as desktop shortcut

Method 3: Standard Mode (Visible Console)

1. Double-click `START_MBAS.bat`
2. Two console windows will appear (backend + frontend)
3. Browser opens automatically
4. Console windows remain visible (useful for debugging)

Login

Default administrator credentials:

```
Username: admin
Password: admin123
```

 **Security Warning:** Change the default password immediately after first login!

Initial Configuration

After first login, configure:

1. **Business Settings** (Settings → Business Info)
 - Business name
 - Address and contact details
 - Tax ID / Registration number
 - Currency (USD, EUR, INR, etc.)
 - Timezone
2. **Upload Business Logo** (Settings → Branding)
 - File formats: PNG, JPG
 - Recommended size: 200x200 pixels
 - Maximum file size: 2 MB
3. **Create User Accounts** (Settings → Users)
 - Create Manager accounts
 - Create Sales User accounts
 - Assign appropriate roles
4. **Configure Invoice Settings** (Settings → Invoice)
 - Invoice prefix (default: INV)
 - Tax rates (GST, VAT, Sales Tax)
 - Payment terms
 - Invoice footer text
5. **Add Initial Product Data** (Inventory → Products)
 - Import products from CSV (optional)
 - Or manually add products

✓ Verification & Testing

Health Check

Run the automated health check:

1. Navigate to MBAS installation folder
2. Double-click `HEALTH_CHECK.bat`
3. Review output:

```
MBAS System Health Check
=====

✓ Python Installation: OK (3.12.x)
✓ Virtual Environment: OK
✓ Backend Dependencies: OK
✓ Database: OK (mbas.db - 2.1 MB)
✓ Frontend Build: OK
✓ System Tray Dependencies: OK

Status: All systems operational ✓
```

Manual Verification

1. Backend API Test

Open browser and navigate to:

```
http://localhost:8000/health
```

Expected response:

```
{
  "status": "healthy",
  "database": "connected",
  "version": "1.0.7"
}
```

2. API Documentation

Access interactive API docs:

```
http://localhost:8000/docs
```

You should see the FastAPI Swagger UI with all endpoints listed.

3. Frontend Test

Navigate to:

```
http://localhost:5173
```

Or:

```
http://localhost:8000
```

You should see the MBAS login page.

4. Database Test

```
cd backend
python -c "from src.core.db import engine; from sqlalchemy import Session; session = Session(engine); print('Database connection: OK')"
```

Expected output:

```
Database connection: OK
```

5. Create Test Invoice

1. Login as admin
2. Go to **Billing** page
3. Add a product to cart
4. Generate invoice
5. Verify invoice number format: `INV-20260428-0001`

6. Test System Tray

1. Right-click system tray icon (green)
2. Verify menu options:
 - Open MBAS
 - Health Check
 - Exit MBAS

Troubleshooting

Installation Issues

Problem: "Python is not recognized"

Cause: Python not added to PATH

Solution:

1. Reinstall Python
2.  Check "Add Python to PATH"
3. Or manually add to PATH:
 - System Properties → Environment Variables
 - Add: `C:\Python312\` and `C:\Python312\Scripts\`

Problem: "pip install failed"

Cause: No internet connection or proxy blocking

Solution:

1. Verify internet connection
2. If behind proxy, configure pip:

```
set HTTP_PROXY=http://proxy.company.com:8080
set HTTPS_PROXY=http://proxy.company.com:8080
pip install -r requirements-lock.txt
```

Problem: "Virtual environment not found"

Cause: Installation incomplete or corrupted

Solution:

1. Delete `venv` folder
2. Run `INSTALL.bat` again

Problem: "Failed to install system tray dependencies"

Cause: No internet connection on first launch

Solution:

1. Connect to internet
2. Click MBAS icon again
3. Dependencies will install automatically

Runtime Issues

Problem: "Failed to start server"

Cause: Port 8000 already in use

Solution:

1. Check if another MBAS instance is running (system tray)
2. Or find and stop conflicting process:

```
netstat -ano | findstr :8000
taskkill /PID <process_id> /F
```

Problem: "Login failed"

Cause: Backend not running or database corrupted

Solution:

1. Verify backend is running: <http://localhost:8000/health>
2. If database corrupted, restore from backup:

```
copy backend\backups\mbas_backup_latest.db backend\mbas.db
```

Problem: "Blank page after login"

Cause: Frontend not loading or JavaScript errors

Solution:

1. Clear browser cache (Ctrl+Shift+Delete)
2. Try different browser
3. Check console for errors (F12 → Console)

Problem: "System tray icon not appearing"

Cause: Tray dependencies missing or failed to install

Solution:

1. Manual installation:

```
venv\Scripts\activate
pip install pystray Pillow psutil
```

2. Restart MBAS

Database Issues

Problem: "Database locked"

Cause: Multiple processes accessing database

Solution:

1. Stop all MBAS instances
2. Run `STOP_MBAS.bat`
3. Delete `backend\mbas.db-wal` and `backend\mbas.db-shm`
4. Restart MBAS

Problem: "Disk I/O error"

Cause: Insufficient disk space or corrupted database

Solution:

1. Free up disk space (minimum 500 MB)
2. Run database integrity check:

```
sqlite3 backend\mbas.db "PRAGMA integrity_check;"
```

3. If corrupted, restore from backup

Performance Issues

Problem: "Slow product search"

Cause: Database needs optimization

Solution:

```
cd backend
python -c "import sqlite3; conn = sqlite3.connect('mbas.db'); conn.execute('VACUUM'); conn.close()"
```

Problem: "High memory usage"

Cause: Large dataset or memory leak

Solution:

1. Restart MBAS regularly (daily for heavy use)
 2. Archive old invoices (move to separate database)
 3. Upgrade RAM to 8 GB if processing > 10,000 products
-



Uninstallation

Complete Removal

1. Stop MBAS:

- Right-click system tray icon → Exit MBAS
- Or run `STOP_MBAS.bat`

2. Backup Data (if needed):

- Copy `backend\mbas.db` to safe location
- Copy `backend\backups\` folder

3. Delete Installation Folder:

```
rmdir /s /q C:\MBAS
```

4. Remove Desktop Shortcut:

- Delete "MBAS" icon from desktop

5. Clean Registry (optional):

- No registry entries created by MBAS
- No system files modified

Data Retention

After uninstallation, you can:

- Keep `mbas.db` file for data archive
- Import into new MBAS installation
- Query using any SQLite viewer



Support & Resources

Documentation

- **User Manual:** `USER_MANUAL.md` (end-user guide)
- **System Preview:** `SYSTEM_PREVIEW.md` (feature overview)
- **API Documentation:** <http://localhost:8000/docs> (when running)

Common Issues

- **Path Fix:** See `PATH_FIX_v1.0.7_FINAL.md`
- **First-Click Reliability:** See `FIRST_CLICK_FIX_v1.0.7.md`
- **Currency Display:** See `CURRENCY_FIX_SUPPLIERS_PURCHASES.md`

Logs

Check logs for detailed error information:

- Backend logs: `backend\logs\app.log`
- Installation logs: `installation.log`

Quick Diagnostics

Run health check anytime:

```
HEALTH_CHECK.bat
```

Or manual check:

```
python -c "import sys; print(f'Python: {sys.version}'); import sqlite3; print(f'SQLite: {sqlite3.sqlite_version}')"

```



Installation Verification Checklist

Use this checklist after installation:

System Verification

- ☐ Python 3.11+ installed and in PATH
- ☐ SQLite3 functional
- ☐ Virtual environment created
- ☐ Backend dependencies installed
- ☐ System tray dependencies installed
- ☐ Desktop shortcut created

Functional Verification

- ☐ MBAS launches without errors
- ☐ System tray icon appears
- ☐ Browser opens automatically
- ☐ Login page loads
- ☐ Can login with admin/admin123
- ☐ Dashboard displays correctly
- ☐ Can create test invoice
- ☐ Can add test product
- ☐ Can run health check

Security Verification

- ☐ Default password changed
- ☐ Additional user accounts created
- ☐ Role-based access working
- ☐ Backup system configured
- ☐ Database file permissions correct

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Next Review: July 2026